

Design of a 2:1 Multiplexer

Implementation on Shrike Lite FPGA

Presented by: Group Electronics Club

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1 The Shrike Lite FPGA

Shrike Lite is an accessible FPGA development platform designed for learning digital logic design. It provides a streamlined environment for mapping Verilog designs onto physical hardware.

Advantages of FPGAs

- **Reconfigurability:** FPGAs can be reprogrammed an infinite number of times to correct logic errors.
- **Parallelism:** FPGAs handle multiple hardware tasks simultaneously, unlike sequential CPUs.
- **Low Latency:** Direct hardware execution ensures minimal delay in signal processing.

Project Overview

This project focuses on the fundamental 2-to-1 Multiplexer (MUX). The MUX selects between two input signals based on a single select line (*sel*). Our goal was to successfully map this logic onto the Shrike Lite fabric.

2 Implementation Code

The Verilog module defines the logic for selecting the output based on the state of the select line.

```
module mux2to1 (
    input wire a,      // Input 0
    input wire b,      // Input 1
    input wire sel,    // Select line
    output wire out     // MUX output
);

assign out = sel ? b : a;
```

Source Code Screenshot

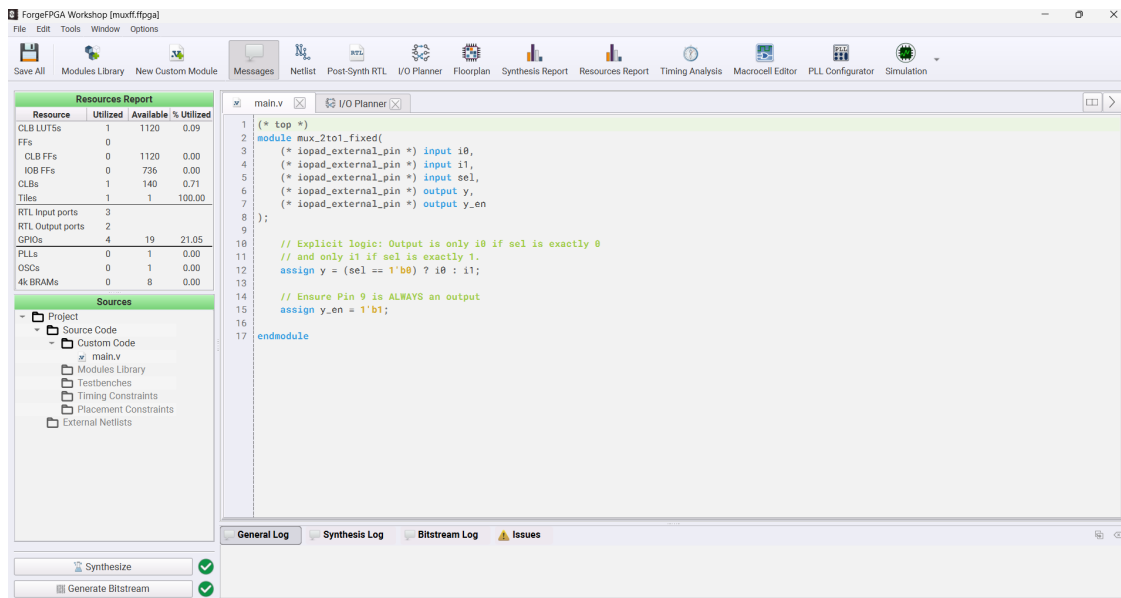


Figure 1: Verilog implementation in the IDE.

3 Hardware Mapping and Results

3.1 IO Planner

The IO Planner defines the physical pin constraints for the inputs and outputs on the board.

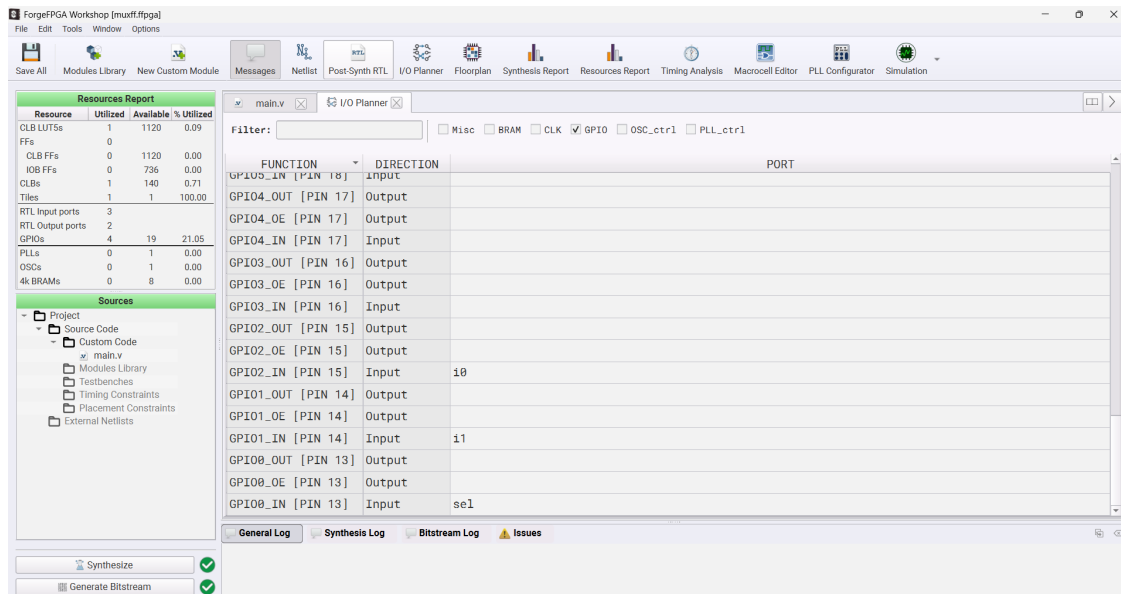


Figure 2: Pin assignment and IO configuration.

3.2 Hardware Verification

Below is the physical result showing the MUX operation on the Shrike Lite board.

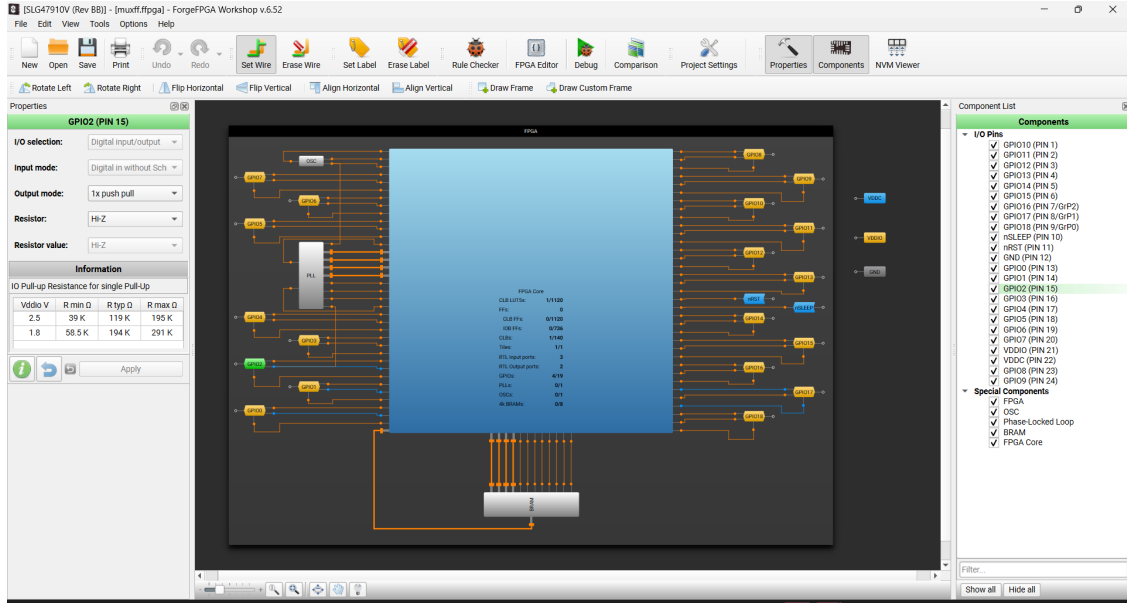


Figure 3: Pin assignment and IO configuration.

4 Simulation Waveform

The functional simulation verifies the logic before hardware deployment. It displays the timing relationship between inputs a, b , the sel line, and the resulting out .

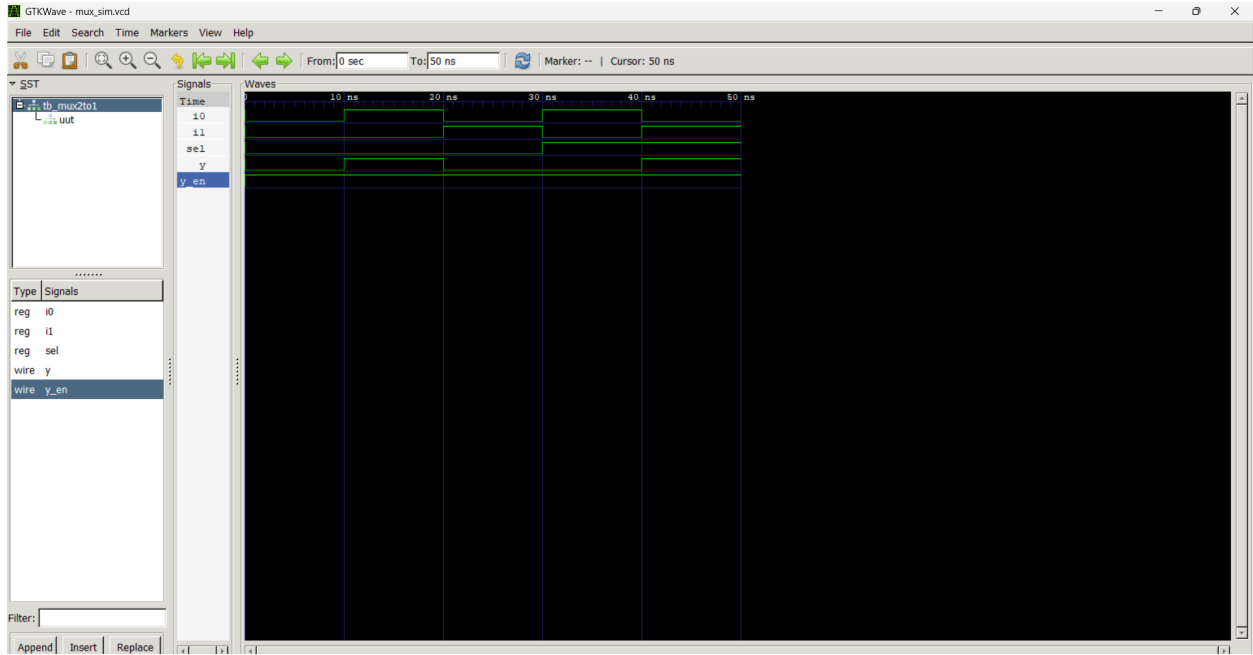


Figure 4: Timing simulation of the 2:1 Multiplexer.

5 Technical Challenges: High-Z and Pull-ups

During the project, we encountered an issue where the select line behaved erratically.

5.1 The Problem: High-Z State

Initially, the select line pin was configured in a **High-Z (High Impedance)** state. In this state, the input is "floating," meaning it is not strictly tied to logic high or logic low. This caused the MUX to pick up environmental noise, leading to flickering outputs.

5.2 The Solution: Pull-up Configuration

To resolve this, we modified the configuration to ensure the select line was never floating.

- We changed the pin constraints to move away from High-Z.
- By utilizing an internal **Pull-up**, we ensured that the select line remained at a stable logic '1' unless actively driven to '0' by the switch.

This change was critical for the reliable operation of the hardware switches on the Shrike Lite board.